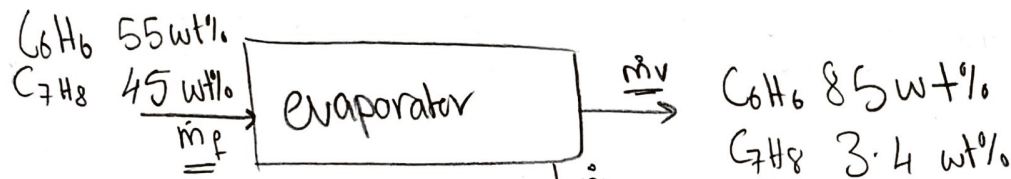
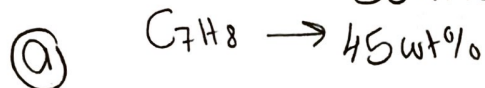
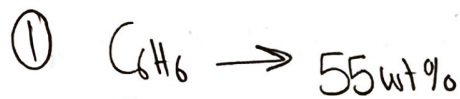


HW3

- No reaction
- St. st.

10.6 wt% residual liquid ✓

③ $Acc = In - out + gen - con$

total $\Rightarrow 0 = m_f - (m_v + m_L)$

$$m_f = m_v + m_L$$

benzene: $0 = 0.55 m_f - 0.85 m_v - 0.106 m_L$

$$0 = 0.55(m_v + m_L) - 0.85 m_v - 0.106 m_L$$

$$0.55 m_v + 0.55 m_L - 0.85 m_v - 0.106 m_L = 0$$

$$-0.3 m_v + 0.444 m_L = 0$$

$$\frac{m_v}{m_L} = 1.48$$

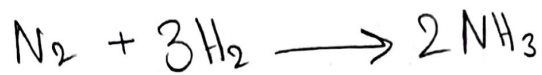
④ underspecified - using the equation given from the book.

3.

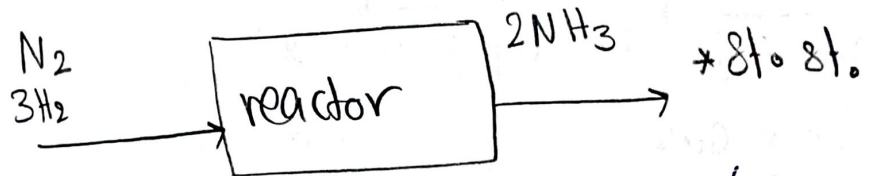
② 1:3 molar ratio to an ammonia synthesis
Nitrogen to hydrogen

69/80

$$R = 0.7302 \frac{\text{ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot \text{R}^\circ}$$



1140°R and 100 atm



The reaction produces 1000 lbmol/h NH₃.
PV = nRT

$$\dot{V}_g = \left(1000 \frac{\text{lbmol}}{\text{h}} * 0.7302 \frac{\text{ft}^3 \cdot \text{atm}}{\text{lbmol} \cdot \text{R}^\circ} * 1140 \text{R}^\circ \right) / 100 \text{ atm}$$

DOF -4.

$$\dot{V}_g = 10514.9 \frac{\text{ft}^3}{\text{h}}$$

* Total volumetric balance:

$$\text{Acc} = \text{In} - \text{out} + \text{gen} - \text{con}$$

$$\text{St. st. } 0 = \dot{V}_i - \dot{V}_e + \dot{V}_g - \dot{V}_c$$

$$\dot{V}_i = ??$$

Nitrogen: $0 = \dot{V}_{\text{N}_2} - 0.75 \dot{V}_{\text{N}_2} + 0 - \dot{V}_{\text{con, N}_2}$

$$\dot{V}_{\text{N}_2} = 0.25 \dot{V}_i$$

$$\frac{\dot{V}_g}{\text{N}_2 \text{ con}} = \frac{2}{1}$$

$$0.25 \dot{V}_i - 0.75(0.25 \dot{V}_i) - \left(\frac{10514.9 \frac{\text{ft}^3}{\text{h}}}{2} \right) = 0$$

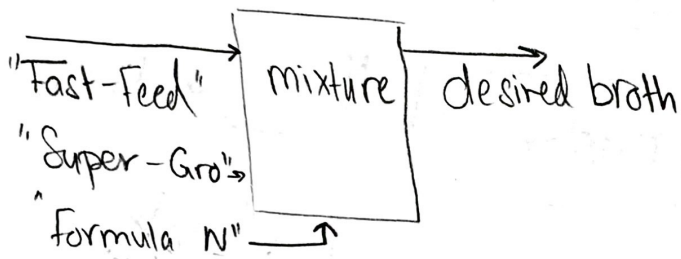
$$\dot{V}_{\text{con}} = \dot{V}_g / 2$$

$$\frac{1}{16} \dot{V}_i = 5257.45 \frac{\text{ft}^3}{\text{h}}$$

$$\dot{V}_i = 8411.92 \frac{\text{ft}^3}{\text{h}}$$

-4.

③



desired broth

15wt% glucose
6 wt% Phosphate
6 wt% nitrate

\$15kg

— assume to produce 1 kg of the desired broth.

$m_F \rightarrow$ mass of Fast feed powder

$m_S \rightarrow$ " " Super-Gro "

$m_N \rightarrow$ " " Formula N "

eq1.) $\frac{0.45 m_F}{0.45} = \frac{0.15 \text{ glucose kg}}{0.45} \Rightarrow m_F = \frac{1}{3} \text{ kg of powder Fast-feed}$

eq2.) $0.02 m_F + 0.35 m_S + 0.12 m_N = 0.06 \text{ kg of phosphate}$ ✓

$0.35 m_S + 0.12 m_N = \frac{4}{75} \Rightarrow m_S = \left(\frac{4}{75} - 0.12 m_N \right) \bigg|_{0.35}$

eq3.) $0.01 m_F + 0.15 m_S + 0.58 m_N = 0.06 \text{ kg of Nitrate}$ ✓

$-0.15 m_S + 0.58 m_N = \frac{17}{300}$ $\frac{0.15}{0.35} \left(\frac{4}{75} - 0.12 m_N \right) + 0.58 m_N = \frac{17}{300}$

$\frac{4}{175} - \frac{9}{175} m_N + 0.58 m_N = \frac{17}{300}$

$\frac{559}{700} m_N = \frac{71}{2100}$

$m_N = \frac{71}{1677} = 0.0423 \text{ kg Formula N}$

$m_S = \frac{1156}{8385} \text{ kg Super-Gro} = 0.138 \text{ kg}$

So 0.33 kg of fast feed and 0.13 kg Super Gro
 And 0.064 kg of Formula N. water that should
 be add is 0.473 kg of H_2O . ✓

$$0.33 + 0.13 + 0.064 = 0.524$$

$$0.524 - 1 = 0.473 \text{ kg of water.}$$

⑥ $0.33 \times \$20 + 0.13 \times \$10 + 0.064 \times \$15 = \8.92

(we are saving): $15 - 8.92 = \$6.08/\text{kg}$

⇒ for 20 kg batch we will be saving $20 \times \$6.08 = \$121/\text{batch}$
 of 20 kg.

* we also need $2000/121 = 16.53$ batch to make up
 for the cost. ✓

⑦ $0.03(0.333) + 0.08(0.13) = 0.020 \text{ kg}$, then the wt% is 2.

Requiring 3wt% trace nutrients likes Def = -1 & the problem
 is overspecified.

* we can increase SG since it is lower in price
 and it has more than 3% of nutrients.

$$0.03(0.333) + 0.08 \text{ SG} = 0.030 \text{ kg}$$

$\text{SG} = 0.25 \text{ kg}$

fast feed can't be decreased
 because it is only the only source
 of glucose.

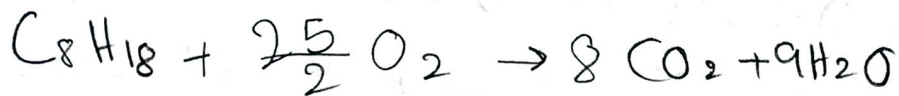
* The quantity of H_2O will decrease to 0.382 kg and the price
 will increase to \$9.66.

P2.60

$$\frac{0.25 \times 5.6 \times 10^9 \text{ metric tons} \times \frac{2200 \text{ lb}}{\text{metric ton}}}{300 \times 10^6 \text{ person}} = 10,300,000 \text{ CO}_2/\text{Person}$$



gasolines combustion



Now, calculating CO_2 emissions per gram (for gasoline)

$$\frac{8 \text{ g mol CO}_2}{\text{g mol C}_8\text{H}_{18}} \times \frac{44 \text{ g/mol}}{114 \text{ g/mol}} = \frac{3.1 \text{ g CO}_2}{\text{g C}_8\text{H}_{18}}$$

Av. person drives 19 miles/day $\Rightarrow$ 13,476 miles/year and gets 27.5 miles/gallon

then each person

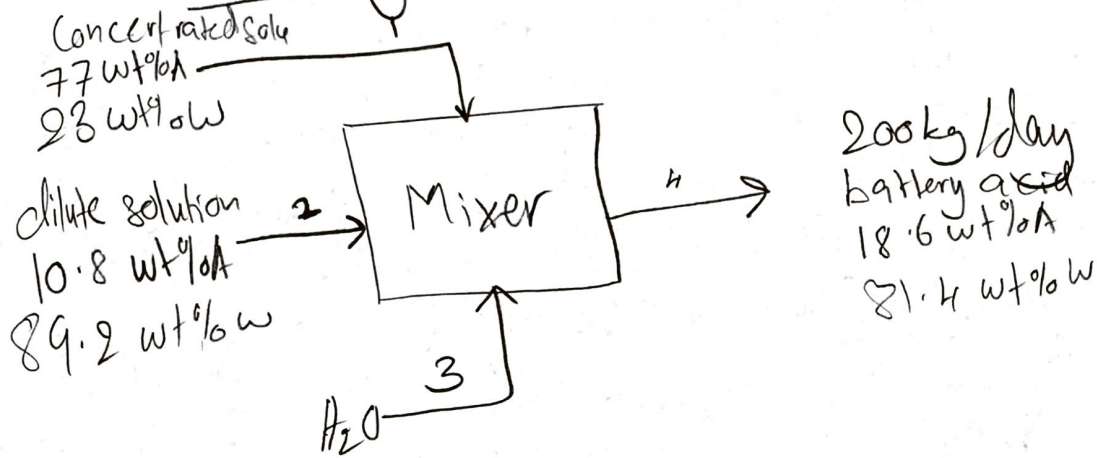
$$13,476 \frac{\text{mi}}{\text{year}} \times \frac{1 \text{ gal}}{27.5 \text{ mi}} \times \frac{1000 \text{ ml}}{0.26417 \text{ gal}} \times \frac{0.7 \text{ g C}_8\text{H}_{18}}{\text{ml}} \times \frac{3.1 \text{ g CO}_2}{\text{g C}_8\text{H}_{18}}$$

$$= 8866.4 \text{ CO}_2$$



* The estimation might be below the real value because we haven't take account public transportation, energy generating.

* Flow Diagram:



DOF = 7 - 6 = 1, The problem is underspecified. ✓

Material balances:

$$\text{Acid: } 0.77 F_1 + 0.108 F_2 = 0.186(200) = 37.2 \text{ kg/day}$$

$$\text{Water: } 0.23 F_1 + 0.892 F_2 + F_3 = 0.814(200) = 162.8 \text{ kg/day}$$

- 2 eqn, 3 unks

$$0.10 F_1 + 0.02 F_2 = \text{cost} (\$/\text{day})$$

$$\text{From } F_1 = 23 \text{ kg/day to } F_1 = 48.3 \text{ kg/day}$$

The minimum cost with the other —

using solver we get

$$m_1 = 48.3 \text{ kg/day}$$

$$m_2 = 151.68 \text{ kg/day}$$

✓